INCH POUND

MIL-I-7077B
2 JULY 1991

SUPERSEDING
MIL-I-7077A
9 DECEMBER 1968

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -70° TO +150°C, 2 INCH SIZE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers electrical resistance and multifunction temperature indicator.

* 1.2 Classification. The indicators shall be of the following part numbers, as specified (see 6.2).

MS28009-1	Fluorescent Dial Markings
MS28009-3	Lusterless White Dial Markings

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Resources and Logistics Services Division, SA-ALC/TIRDM, Kelly AFB, TX 78241-5609, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6685
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Military

MIL-P-116	Preservation, Methods of
MIL-I-7070	Indicator, Temperature, Electrical-Resistance Hermetically Sealed, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MS28009	Indicator-Temperature, Electrical Resistance Multifunction, -70° to +150°C, 2 Inch Size
MS33558	Numerals and Letters - Aircraft Instrument Dial, Standard Form of
MS33639	Cases, Instrument, Clamp Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Official Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, IL 60606.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Qualification. The indicators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award on contract (see 4.5 and 6.3).

* 3.2 General specification. The requirements of MIL-I-7070 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.2.1 In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.3 Construction.

3.3.1 Case. The case shall be in accordance with MS33639 as shown on MS28009.

3.3.2 Scale length. The scale of the indicator shall be as long as practicable and shall not be less than 1-3/4 inches as measured on the outer arc of the shortest graduation.

3.3.3 Pointer. The pointer design shall be as shown on figure 1.

3.3.3.1 Pointer overswing. The pointer overswing shall not be less than 3 percent of full scale angular deflection beyond each end of the scale.

3.4 Marking. The markings shall be durable and in accordance with figure 1. The lusterless white shall be color 37875 of FED-STD-595. The lusterless black shall be color 37038 of FED-STD-595.

3.5 Electrical receptacle. An electrical receptacle, MS3367B-14S-2P, shall be provided and located as shown on MS28009. Receptacle terminals shall be identified as follows:

Terminal	Connection
A	+24 volt battery
B	Resistance bulb
C	Common-negative 24 volt battery and remaining resistance bulb connection
D	Open

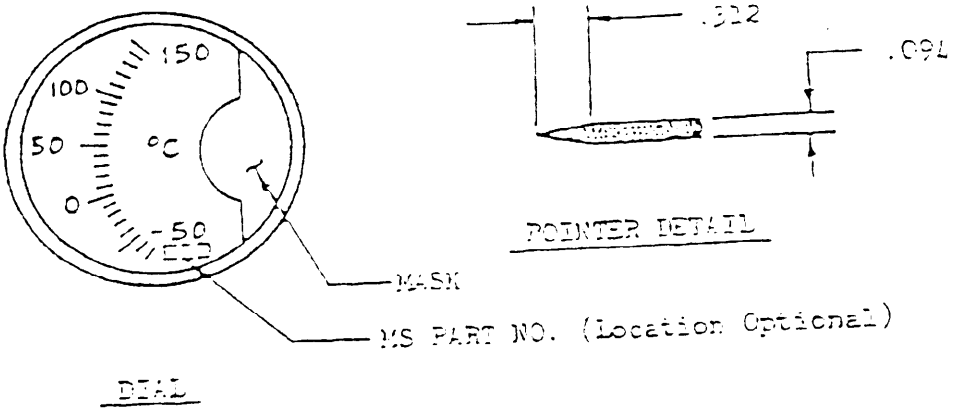
3.6 Weight. The weight of the complete indicator shall not exceed 1.4 pounds.

3.7 Identification.

3.7.1 Identification of products. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.7.2 Instrument identification decalcomanias. The indicator shall be provided with 1 each of the 2 identification decals as shown on figure 2.

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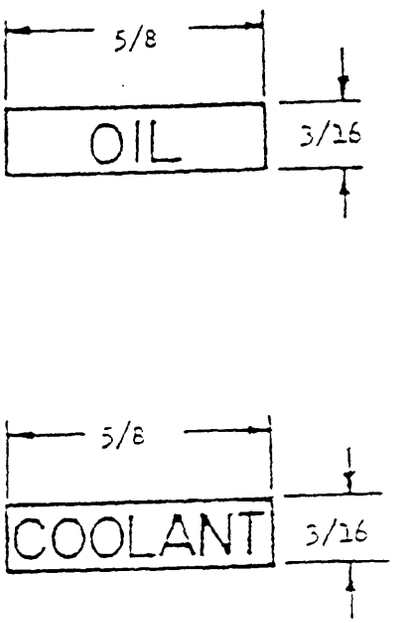


MARKING	HEIGHT OR LENGTH ± 0.005	WIDTH OF LINE OR GRADUATION ± 0.005	MATERIAL OR FINISH
ALL NUMERALS MAJOR GRADUATIONS MINOR GRADUATIONS LETTERING "°C" SHADED PORTION OF POINTER	0.125 0.156 0.093 0.125 -----	0.020 0.031 0.020 0.020 -----	FLUORESCENT LUMINESCENT MATERIAL OR LUSTERLESS WHITE
BACKGROUND OF DIAL (MS PART NO.)	----- 0.062	----- 0.010	LUSTERLESS BLACK

DIMENSIONS IN INCHES.

FIGURE 1. Dial, Pointer and Markings

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DIMENSIONS IN INCHES.

UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/32.

FIGURE 2 - DECAL SIZE AND MARKING

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tters shall be finished lusterless white to correspond to the dial finish. ckground of the decal shall be transparent or lusterless black. The letters be 0.125 ±0.010 inch high and the width of the lines shall be 0.020 ±0.005 The form of the letters shall conform to MS33558. The decal shall be of the applied type and shall be suitable for application to the cover glass of the tor. The decal when mounted on the cover glass of the indicator, shall be e of meeting the tests specified in Section 4 of this specification without g, blistering, fading, loss of adhension, cracking, softening or hardening. calcomanias, 4 hours after application, shall adhere sufficiently to the so that the surface shall withstand normal handling. Each set of decals be packaged in an envelope marked "THIS ENVELOPE CONTAINS IDENTIFICATION " Complete and detail instructions for mounting the decal on the cover glass indicator shall be furnished with each set of decal.

7.3 Recycled, reclaimed, and virgin materials. Except when intended use of em will be jeopardized by the use of reclaimed or recycled materials, in ing new and revising or amending specifications, preparing activities will that:

There is no exclusion to the use of recovered materials.
There is no requirement that an item be manufactured from virgin materials.
Within 1 year from the date of issue by the Environmental Protection Agency of guidelines designating items which are or can be produced with recovered materials, specifications for such products require the use of recovered materials to the maximum extend possible.

QUALITY ASSURANCE PROVISIONS

1 Responsibility for inspection. Unless otherwise specified in the contract chase order, the supplier is responsible for the performance of all tion requirements as specified herein. Except as otherwise specified, the er may utilize his own facilities or any commercial laboratory acceptable to vernment. The Government reserves the right to perform any of the tions set forth in the specification where such inspections are deemed sary to assure supplies and services conform to prescribed requirements.

1.1 Responsibility for compliance. All indicators must meet all ements of sections 3 and 5. The inspection set forth in this specification become a part of the contractor's overall inspection system or quality m. The absence of any inspection requirements in the specification shall not e the contractor of the responsibility of assuring that all products or es submitted to the Government for acceptance comply with all requirements of ntract. Sampling in quality conformance does not authorize submission of defective material, either indicated or actual, nor does it commit the ment to acceptance of defective material.

2 Classification of inspection. The inspection requirements specified are classified as follows:

a. Qualification inspection (4.5)

b. Quality conformance inspection (4.6)

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c. Inspection of the perservation, packaging, packing and marking for shipment and storage (5.1)

4.3 Test conditions.

4.3.1 Test voltage. The test voltage shall be 28.5 ±0.5 volts direct current (dc), unless otherwise specified.

4.4 Test tolerances. The tolerances specified in table I, not enclosed in parenthesis, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in table I, enclosed in pin parenthesis, are applicable only when the indicators are subjected to a specific test independent of sequence.

* 4.5 Qualification tests. The qualification tests shall consist of all the tests specified herein and in MIL-I-7070.

* 4.6 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests.

b. Sampling tests.

* 4.6.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.7.

a. Scale error at room temperature.

b. Position error.

c. Friction error.

d. Insulation leakage.

* 4.6.2 Sampling tests.

* 4.6.2.1 Sampling plan A.

a. Individual tests.

b. Reduced voltage operation.

c. Magnetic effect.

d. Resistance bulb current.

e. Vibration error.

* 4.6.2.2 Sampling plan B.

a. Sampling plan A.

b. Response time.

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TABLE I

TEST TOLERANCE

Test Point	Test Resistance	Scale Error at Room Temperature	Reduced Voltage	Scale Error at Low Temperature	Scale Error at High Temperature	Scale Error at Room Temperature After Open and Short Circuit	Scale Error at Room Temperature After Vibration Failure
°C	Ohms	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C
-70	60.27	3.0 (3.0)	2.0 (2.0)	7.0 (7.0)	7.0 (7.0)	10.0 (4.0)	10.0 (7.0)
-30	80.56	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
0	90.38	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	10.0 (3.0)	10.0 (6.0)
+30	100.91	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
80	120.36	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
120	137.78	3.0 (3.0)	3.0 (3.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
150	151.91	3.0 (3.0)	3.0 (3.0)	7.0 (7.0)	7.0 (7.0)	10.0 (4.0)	10.0 (7.0)

Test Point	Test Resistance	Extreme Low Temperature Exposure		Scale Error at Room Temperature After High Temperature Exposure	Scale Error with Bulb Lead Resistance
		Scale Error at Low Temperature	Scale Error at Room Temperature After Low Temperature Exposure		
°C	Ohms	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C
-70	60.27	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
-30	80.56	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
0	90.38	12.0 (4.0)	10.0 (2.0)	10.0 (2.0)	2.0 (2.0)
+30	100.91	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
80	120.36	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
120	137.78	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
150	151.91	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)

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c. Damping factor.

d. Power consumption.

e. Bulb lead resistance.

* 4.7 Test methods.

* 4.7.1 Scale error at room temperature. The test resistance specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in table I shall not exceed the tolerances specified in table I.

* 4.7.2 Position error. When the indicator is subjected to the position error test, the maximum change in indication shall not exceed 4° Centigrade (C).

* 4.7.3 Friction error. When the indicator is subjected to the friction test, the maximum difference in reading before and after tapping shall not exceed 4°C.

* 4.7.4 Insulation leakage. When the indicator is subjected to the insulation leakage test, the insulation resistance shall be a minimum of 20 megohms.

* 4.7.5 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5 volts at the test points specified in table I. The indications at the reduced voltage shall not differ from the indications with 28.5 volts applied by more than the tolerances specified in table I.

* 4.7.6 Magnetic effect. When the indicator is subjected to the magnetic effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

* 4.7.7 Resistance bulb current. A total resistance of 100 ohms shall be applied across the bulb circuit terminals of the indicator, the test voltage applied, and the bulb circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 milliampere (ma).

* 4.7.8 Vibration error. When the indicator is subjected to the vibration error test, the maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 4°C. The maximum difference between the original position of the pointer during vibration shall not exceed 4°C. Pointer oscillation shall not be considered as pointer variation.

* 4.7.9 Response time. When the indicator is subjected to the response time test, the response time shall not exceed 3 seconds.

* 4.7.10 Damping factor. When the indicator is subjected to the damping factor test, the damping factor shall not be less than 1.6.

* 4.7.11 Power consumption. When the indicator is subjected to the power consumption test, the power consumption shall not exceed 1.5 watts.

* 4.7.12 Bulb lead resistance. The indicator shall be connected as shown on figure 3 and shall be subjected to a scale error at room temperature test. The

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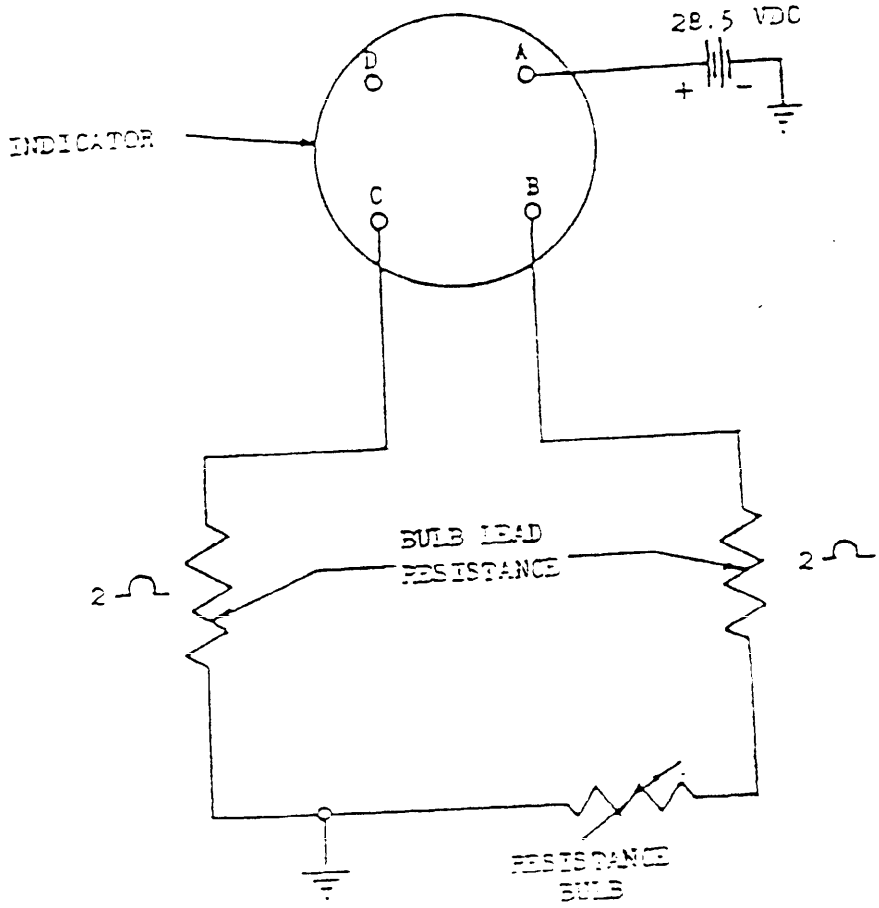


FIGURE 3 - WIRING DIAGRAM

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indicator with bulb lead resistance shall not differ from the indication without bulb resistance, by more than the tolerances specified in table I.

* 4.8 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2d).

5.1.1 Level A. Unless otherwise specified in the contract or order, the indicators shall be individually preserved and packaged in accordance with Method III of MIL-P-116 in a unit container conforming to type CF, class Domestic Variety SW, grade 125 of PPP-B-636.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's standard practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Indicators, preserved and packaged as specified in 5.1.1, shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas type containers conforming to PPP-B-601. The gross weight of each container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size, and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-118o.

5.2.2 Level B. Indicators, preservec and packaged as specified in 5.1.1, shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Cusioning, blocking and bracing shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

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5.4 Marking. In addition to any other marking required by order or contract, the interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The electrical resistance temperature indicators covered by this specification are intended for use in aircraft to indicate cabin or free air temperature.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of specification.

b. Designate indicator desired (see 1.2).

c. Ordering data in accordance with MIL-I-7070.

d. Levels of preservation, packaging and packing required (see 5).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products with are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-7077) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts of purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is SA-ALC/TIRDM, Kelly AFB, TX 78241, and information pertaining to qualification of products may be obtained from that office.

* 6.4 Bezel. The temperature indicator covered by this specification can be used on instrument panels requiring MS33638 case mounting by the application of an instrument mounting bezel in accordance with MS28055.

* 6.5 Subject term (key word) listing.

Fluorescent Dial Marking, Indicator
Indicator
Indicator, Temperature
Indicator, Temperature, Electrical Resistance
Indicator, Temperature, 2 inch size
Indicator, Temperature, Fluorescent Dial Marking
Lusterless White Dial Marking, Indicator
Lusterless Indicator

* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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Custodian:
Air Force - 99

Review activities:
Air Force - 11
Army - ME
Navy - WP

Preparing activity:
Air Force - 82

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